Laboratory Safety Handbook

Marlow Research Institute · Second edition, valid from 1 September 2026

1 Introduction

1.1 Purpose

This handbook sets out the rules that keep everyone safe in the institute's laboratories. It applies to staff, students and visitors from the moment they enter a laboratory.

1.2 Scope

The rules cover three areas of daily work.

- Personal protective equipment
- Handling, storage and disposal of chemicals
- Emergency procedures and first aid

> Never work alone in a laboratory after 18:00.

2 Personal protective equipment

2.1 Required equipment

The table shows which equipment is required for each type of task.

Task	Lab coat	Gloves	Goggles	Face shield
Weighing solids	Required	Required	Optional	—
Handling acids and bases	Required	Required	Required	Optional
Working with solvents	Required	Required	Required	—
Using the autoclave	Required	Heat-resistant	Required	Required

2.2 Care of equipment

Protective equipment must be checked before every use and replaced when it is damaged.

1. Check the equipment before use
    a. look for cracks, holes and stains
    b. check that goggles seal against the face
2. Store coats and goggles outside the laboratory
3. Report damaged equipment to the safety officer

3 Chemical handling

3.1 Storage

Chemicals are stored by hazard class and never alphabetically. Flammable liquids belong in the ventilated cabinet next to the door.

3.2 Spills

> Do not try to clean up a spill of an unknown substance.

In case of a small spill, proceed as follows:

1. Warn the people around you and stop the work
2. Put on gloves and goggles, then
    a. cover the spill with absorbent material
    b. sweep the material into a labelled bag
3. Wash the area with water and soap

3.3 Waste

Waste stream	Container	Examples
Halogenated solvents	Red can with vent cap	Chloroform, dichloromethane
Other organic solvents	Blue can	Ethanol, acetone, hexane
Sharps	Yellow puncture-proof box	Needles, broken glass
Aqueous solutions	White canister	Buffers, dilute acids and bases

4 Emergencies

4.1 Contacts

Fire and ambulance	112
Institute security (24 hours)	ext. 5000
Safety officer	ext. 5120
Poison information	0800 555 0199

4.2 Evacuation

When the alarm sounds, leave the laboratory immediately and close the door behind you. Do not use the lifts, and do not return until the safety officer allows it.

> Gather at the assembly point in the east car park and report to your group leader.

5 Document control

This handbook is reviewed every year by the safety committee. Printed copies are not controlled; the current version is on the intranet.

Edition	Date	Change
1st	March 2023	First issue
2nd	September 2026	New waste streams, updated contacts